

MRI-Based Diagnosis and Multi-Class Classification of Alzheimer's Disease Stages

Paper Overview

Topic: Automated Multi-Class Alzheimer's Disease Classification using Enhanced MRI Super-Resolution and Lightweight Deep Neural Networks

This summary examines a state-of-the-art study on the use of structural magnetic resonance imaging (MRI) for classifying Alzheimer's disease (AD) across four clinical stages. It evaluates an end-to-end computer vision pipeline integrating contrast-limited histogram equalization, generative adversarial super-resolution enhancement, and lightweight feature extraction—spanning from raw image preprocessing to automated multi-class diagnosis.

Paper Title	Author(s) / Year	Dataset / Source	Model / Technique	Key Findings
A Hybrid Deep Learning Multi-Class Classification Model for Alzheimer's Disease Using Enhanced MRI Images	Alwakid (2026)	Augmented Alzheimer MRI Dataset (Kaggle): 4 classes (Non-Demented, Very Mild Demented, Mild Demented, Moderate Demented).	CLAHE (Contrast Limited Adaptive Histogram Equalization) + ESRGAN (Enhanced Super-Resolution GAN) + MobileNetV2 Architecture.	Achieved 98.88% overall classification accuracy and a 98.00% Macro F1-score . Super-resolution preprocessing significantly boosts detection accuracy for subtle early-stage dementia (Very Mild Demented).

Summary

The analyzed study converges on three core technical layers needed to build an effective MRI-based multi-class diagnostic system for Alzheimer's Disease:

- Contrast Enhancement & Super-Resolution Preprocessing:** Standard structural MRI scans often lack sufficient contrast to expose subtle structural variations in early-stage dementia. The integration of **CLAHE** with **ESRGAN** effectively sharpens tissue contrast and reconstructs high-resolution spatial details around ventricular enlargement and cortical atrophy regions.

2. **Lightweight Feature Extraction:** Instead of using heavy, computationally expensive networks, the study utilizes **MobileNetV2** as the core feature extraction engine. Applying depthwise separable convolutions to super-resolved input scans allows the system to capture complex neurodegenerative patterns without high computational complexity.
3. **Multi-Stage Disease Quantification:** The primary strength lies in its capacity to reliably discriminate between four distinct severity stages (Non-Demented, Very Mild Demented, Mild Demented, and Moderate Demented), achieving a peak classification accuracy of **98.88%** and an **F1-score of 98.00%**.

Limitations & Gaps:

The primary remaining limitation in this work is the reliance on heavy GAN-based preprocessing (ESRGAN), which introduces high computational latency, as well as the absence of explicit spatial or channel-wise attention mechanisms within the core CNN backbone to directly visualize localized brain tissue atrophy.